

Math 141, S06
Quiz 0

Name:
SSN: xxx-xx-

Instructions: There are totally 4 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. Simplify

$$\frac{1}{3} + \frac{3}{4} = \frac{4}{12} + \frac{9}{12} = \frac{13}{12}$$

Problem 2. Find

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\tan 45^\circ = 1$$

Problem 3. Simplify

$$x^6 * x^{\frac{1}{3}} = x^{6+\frac{1}{3}} = x^{\frac{19}{3}}$$

Problem 4. Solve the following quadratic equation:

$$x^2 + 2x - 3 = 0.$$

Solution:

$$\begin{aligned} x^2 + 2x - 3 &= 0 \\ (x + 3)(x - 1) &= 0 \end{aligned}$$

So we have

$$x = -3 \quad \text{or} \quad x = 1$$